CAS CS 538. Solutions to Problem Set 6

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How to submit: See instructions on Problem Set 1.

Problem 1. (30 points total) Let F be a secure PRF with n-bit strings for keys, inputs, and outputs.

(a) (5 points) Let $F_1(k, x)$ computed as follows: compute $F(k, x)$, append the last bit of x, and output the result. Prove that F_1 is insecure.

Solution. The main idea behind this proof is the fact that the function, F_1 exposes a part of the input in its output. This can be exploited.

Making the Game

Let $\mathcal{A}$ be the adversary that will play the game of distinguinish F_1 from just some rnd function.

- $\mathcal{A}$ chooses any $x \in^R \{0,1\}^n$
- Push x into Oracle, producing $y \in \{0,1\}^{n+1}$
- Now we check if $x[-1] = y[-1]$
  - If YES, $\hat{b} = 1$
  - Else, $\hat{b} = 0$
- Return $\hat{b}$ to challenger.

Analysis

Let $\text{PRFadv}[\mathcal{A}, F_1] = |\Pr[W_0] - \Pr[W_1]|$ be the magnitude of difference of $\mathcal{A}$ outputting 1 in either experiments $0, 1$.

$\exp 0$**: Using** F_1 Assume that the challenger chooses $k \in^R \{0,1\}^n$. In this experiment, the oracle is F_1. Then, by definition, the last bit of the output will ALWAYS be the last bit of x. Therefore $\hat{b} = 1$ always occurs.

$\exp 1$**: Using rnd** Now, the oracle is just a rnd func R. This means that every bit of y is uniformly distributed and random to x's bits. Thus, the chance of $\hat{b} = 1$ is $\frac{1}{2}$.

Adv

Thus, the advantage for this game can be calculated as:

$$\text{PRFadv}[\mathcal{A}, F_1] = |1 - \frac{1}{2}| = \frac{1}{2}$$

This is non-negligible, thus meaning that F_1 is insecure. $\square$

(b) (5 points) Let $F_2(k, x)$ computed as follows: compute $F(k, x)$, append the last bit of k, and output the result. Prove that F_2 is insecure.

Solution. This issue now is that the key is exposed in the output, which can be exploited.

Game

$\mathcal{A}$ is playing to distinguish F_2 from a rnd function.

- $\mathcal{A}$ chooses $x_1, x_2 \in^R \{0,1\}^n$
- Push x_1, x_2 to Oracle to get $y_1, y_2 \in \{0,1\}^n$
- Now, check if $y_1[-1] == y_2[-1]$. This will set $\hat{b}$
  - YES $\rightarrow 1$
  - ELSE $\rightarrow 0$
- Return $\hat{b}$ to $\mathcal{A}$

Analysis

Let $\text{PRFadv}[\mathcal{A}, F_2] = |\Pr[W_0] - \Pr[W_1]|$ be the magnitude of difference of $\mathcal{A}$ outputting 1 in either experiments $0, 1$.

$\exp 0$**: Using** F_2 Assume that the challenger chooses $k \in^R \{0,1\}^n$. Denote k_{last} to be the last bit of k. In this experiment, the oracle is F_2. Then, by definition, the last bit of outputs y_i will ALWAYS equal to k_{last}. Therefore $\hat{b} = 1$ always occurs.

$\exp 1$**: Using rnd** Now, the oracle is just a rnd func R. This means that every bit of y_i is uniformly distributed and random to k's bits. Thus, the chance of $\hat{b} = 1$ is $\frac{1}{2}$.

Adv

Thus, the advantage for this game can be calculated as:

$$\text{PRFadv}[\mathcal{A}, F_2] = |1 - \frac{1}{2}| = \frac{1}{2}$$

This is non-negligible, thus meaning that F_2 is insecure. $\square$

(c) (10 points) Define F_3 for $2n$-bit keys, $2n$-bit inputs, and $2n$-bit outputs, as follows:

$$F_3((k_1, k_2), (x_1, x_2)) = F(k_1, x_1)||F(k_2, x_2)\,,$$

where $||$ denotes concatenation of strings. Prove that F_3 is insecure.

Solution. The main issue is that parts of the output are dependent on PARTS of the input. I can change just 1 part of the input, and not completely change the output.

Game

$\mathcal{A}$ is playing to distinguish F_3 from a rnd function.

- $\mathcal{A}$ chooses $x_1, x_2, x_2' \in^R \{0,1\}^n$, where $x_2 \neq x_2'$
- Push (x_1, x_2) to Oracle to get $y = y_1||y_2$, with each $y_i \in \{0,1\}^n$
- Push (x_1, x_2') to Oracle to get $y' = y_1'||y_2'$, with each $y_i' \in \{0,1\}^n$
- Now, check if $y_1 == y_1'$. This will set $\hat{b}$
  - YES $\rightarrow 1$
  - ELSE $\rightarrow 0$
- Return $\hat{b}$ to $\mathcal{A}$

Analysis

Let PRFadv$[\mathcal{A}, F_3] = |\Pr[W_0] - \Pr[W_1]|$ be the magnitude of difference of $\mathcal{A}$ outputting 1 in either experiments $0, 1$.

$\exp 0$**: Using** F_2 Assume that the challenger chooses $k_1, k_2 \in^R \{0,1\}^n$. In this experiment, the oracle is F_3. Note how, in both outputs of y, we're using x_1. That then means that both y_1, y_1' are dependent on x_1. Therefore, it should be the case that $y_1 = y_1'$, and $\hat{b} = 1$ always occurs.

$\exp 1$**: Using rnd** Now, the oracle is just a rnd func R. This then means that both outputs of both y's bits are uniformly distributed and random to ANY of x_i's bits. Thus, the chance of $\hat{b} = 1$ occuring is $\frac{1}{2^n}$.

Adv

Thus, the advantage for this game can be calculated as:

$$\text{PRFadv}[\mathcal{A}, F_2] = |1 - \frac{1}{2^n}| \approx 1$$

$\frac{1}{2^n}$ approaches negligiblity as n increases. A negligible minus a non-negligible is non-neglibile (the result is a constant, in simpler terms). Thus, this is non-negligible, meaning that F_3 is insecure. $\square$

(d) (10 points) Define F_4 for $2n$-bit keys, $2n$-bit inputs, and n-bit outputs, as follows:

$$F_4((k_1, k_2), (x_1, x_2)) = F(k_1, x_1) \oplus F(k_2, x_2)\,.$$

Prove that F_4 is insecure.

Solution. The issue, like before, is the fact that sections of the output are dependent on sections of the input, allowing an adversary to modify inputs to find a pattern. It's a bit harder for this case tho, but XORs can be cancelled out.

Game

$\mathcal{A}$ is playing to distinguish F_4 from a rnd function.

- $\mathcal{A}$ chooses $x_1, x_2, x_1', x_2' \in^R \{0,1\}^n$, where $x_1 \neq x_1', x_2 \neq x_2'$
- Push the following 4 inputs into Oracle to get the following $y_i \in \{0,1\}^n$:
  - $(x_1, x_2) \to$ Oracle $\to y_1$
  - $(x_1, x_2') \to$ Oracle $\to y_2$
  - $(x_1', x_2') \to$ Oracle $\to y_3$
  - $(x_1', x_2) \to$ Oracle $\to y_4$
- Now, check if $y_1 \oplus y_2 \oplus y_3 \oplus y_4 = \{0\}^n$. This will set $\hat{b}$
  - YES $\to 1$
  - ELSE $\to 0$
- Return $\hat{b}$ to $\mathcal{A}$

Analysis

Let PRFadv$[\mathcal{A}, F_4] = |\Pr[W_0] - \Pr[W_1]|$ be the magnitude of difference of $\mathcal{A}$ outputting 1 in either experiments $0, 1$.

$\exp 0$**: Using** F_2 Assume that the challenger chooses $k_1, k_2 \in^R \{0,1\}^n$. In this experiment, the oracle is F_4. Note the number of times each variables x_1, x_2, x_1', x_2' appears in the equation of $y_1 \oplus y_2 \oplus y_3 \oplus y_4$ ($y_i = (x_{\text{smth}}) \oplus (x_{\text{smth}})$). What you'll find is that, each of the following variable I have just listed out will come out EXACTLY TWICE.

Also note how many times k_1, k_2 appears in the same equation. As we're XORing 4 terms, these 2 variables appear an EVEN NUMBER of times. By the properties of XOR, this then means that $y_1 \oplus y_2 \oplus y_3 \oplus y_4$ entirely cancels out to 0. Therefore, $\hat{b} = 1$ always occurs.

$\exp 1$**: Using rnd** Now, the oracle is just a rnd func R. This means that each bit of each particular y_i will be uniform and random to one another. However, I am not necessarily saying that each y_i needs to have the same exact bit. $y_1 \oplus y_2 \oplus y_3 = z$ is random, so you could just simlify to check if $z \oplus y_4 = \{0\}^n$. As we're just checking if XORing a rnd value yields all 0s, the chance of this happening is $\frac{1}{2^n}$.

Adv

Thus, the advantage for this game can be calculated as:

$$\text{PRFadv}[\mathcal{A}, F_4] = |1 - \frac{1}{2^n}| \approx 1$$

$\frac{1}{2^n}$ approaches negligiblity as n increases. A negligible minus a non-negligible is non-neglibile (the result is a constant, in simpler terms). Thus, this is non-negligible, meaning that F_4 is insecure. □

Problem 2. (20 points) There are many PRF designs out there based on different assumptions. Suppose you aren't sure which ones are really secure and want to hedge your bets. In this problem, you will prove that you can take multiple different PRF designs and combine them so that the result is secure as long as at least one them is secure. This type of risk reduction is called a PRF *combiner*. There are also combiners for other cryptographic primitives, like CPA-secure encryption, collision-resistant hashing, etc. In this problem, you will prove that XOR is a good PRF combiner.

Let F_1 and F_2 be functions with n-bit keys, inputs, and outputs. Let $F((k_1, k_2), x) = F_1(k_1, x) \oplus F_2(k_2, x)$. Prove that if either F_1 or F_2 is a secure PRF, then F is a secure PRF. (In fact, it suffices to prove that if F_1 is secure than F is secure, because the proof for F_2 is the same. So just prove it for F_1 to get full credit.)

NB: your proof will, of course, proceed via a reduction. In your reduction, be sure to analyze two cases: when the F_1 adversary is in Experiment 0 of the PRF Attack Game (Game 4.2 in Section 4.4.1), you need to prove that the F adversary is also seeing Experiment 0; when the F_1 adversary is in Experiment 1, you need to prove that the F adversary is also seeing Experiment 1.

Solution. Your solution goes here

Problem 3. (20 points, at 10 each)

(a) Let $F : \mathcal{K} \times \mathcal{X} \to \mathcal{Y}$ with $\mathcal{K} = \mathcal{X} = \mathcal{Y} = \{0,1\}^\lambda$ be a secure PRF. Define $F^1(k, x)$ is follows: if $k = 0^\lambda$, output 0^λ; else output $F(k, x)$. Prove that F^1 is secure. Hint: use the Difference Lemma (Theorem 4.7 in Section 4.4.3).

Solution. Your solution goes here

(b) (10 points) Use the previous part to prove the following. Let $H : \mathcal{K} \times \mathcal{X} \to \mathcal{Y}$ with $\mathcal{K} = \mathcal{X} = \mathcal{Y} = \{0,1\}^\lambda$ be a secure PRF. Define $H^1(k', x')$ as follows: set $k = x'$ and $x = k'$; output $H(k, x)$ (that is, simply swap the key and the output). Prove that H^1 is not necessarily a secure PRF.

Solution. Your solution goes here

Problem 4. (30 points, at 15 each) Problem 5.3 from Section 5.8 of the Boneh + Shoup textbook. Specifically,

(a) Assume that $\mathcal{E}$ is CPA-secure and prove that $\mathcal{E}$ is Alt-CPA-secure.

(b) Assume that $\mathcal{E}$ is Alt-CPA-secure and prove that $\mathcal{E}$ is CPA-secure.

Hint 1: Do these parts one a time, to avoid getting lost. Part (b) requires a hybrid argument with many hybrid points.

Hint 2: The textbook presents the Alt-CPA definition as a single randomized experiment, with the bit b chosen at random by the challenger, and the adversary is trying to guess b by outputting a matching $\hat{b}$. This definition style is called "bit-guessing"; define

$$\text{Alt-CPAadv}^*(\mathcal{A}, \mathcal{E}) = |\Pr[b = \hat{b}] - 1/2| \,.$$

This definition is equivalent to a two-experiment definition that we are more familiar with: one for $b = 0$, and the other for $b = 1$. In this view, there is no random choice of b, and we are not measuring whether $b = \hat{b}$; rather, we measure the probability difference between $p_0 = \Pr[\hat{b} = 1]$ in Experiment 0 and $p_1 = \Pr[\hat{b} = 1]$ in Experiment 1; define

$$\text{Alt-CPAadv}(\mathcal{A}, \mathcal{E}) = |p_1 - p_0| \,.$$

The two definitions are equivalent; this equivalence is shown in Section 2.2.5 in the textbook (in the context of semantic security, but it's the same for all other such definitions). It is easier to use CPAadv* / Alt-CPAadv* for part (a), and CPAadv / Alt-CPAadv for part (b).

Solution. Your solution goes here